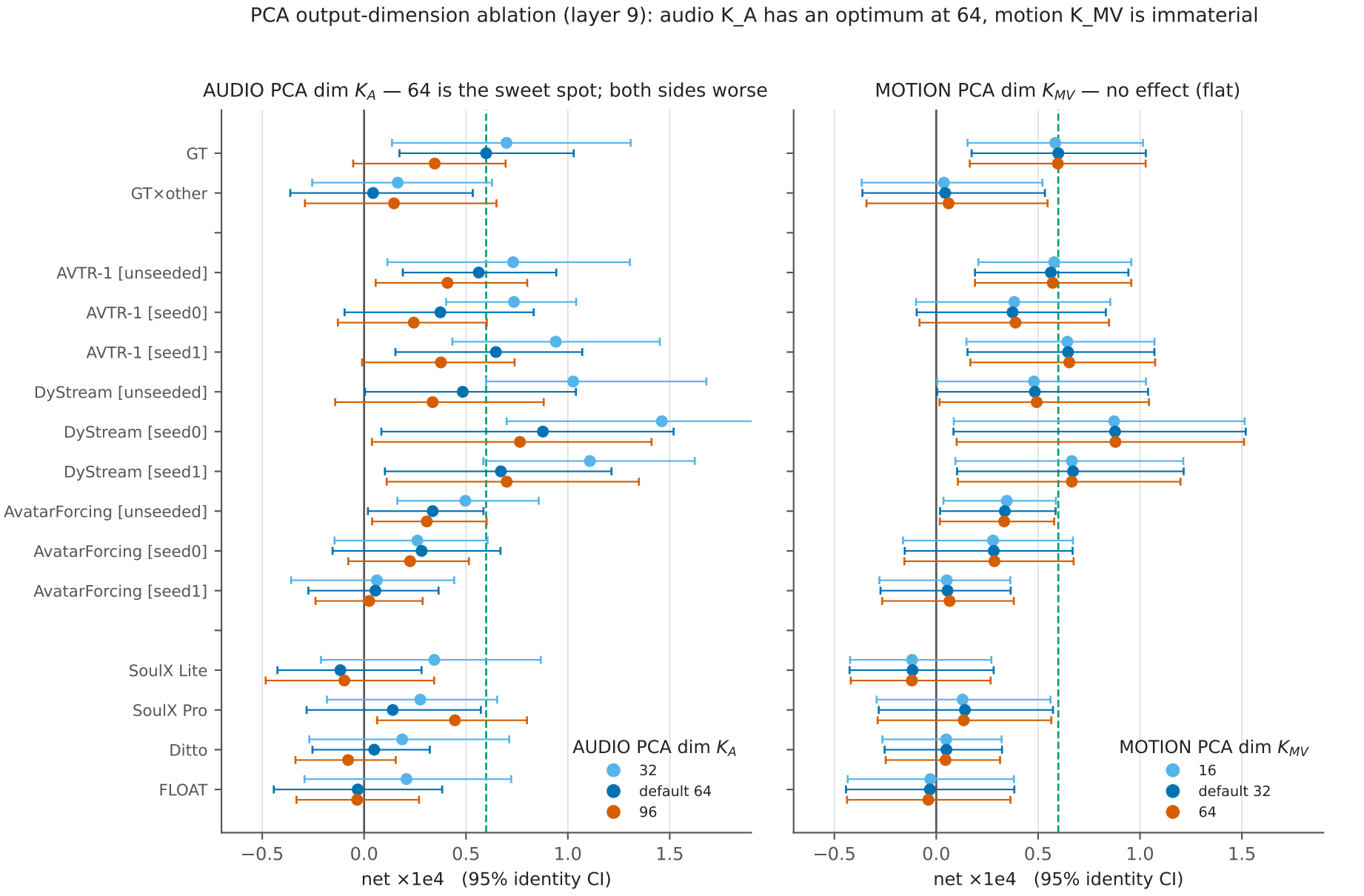
PCA output-dimension ablation (layer 9): audio K_A has an optimum at 64, motion K_MV is immaterial
AUDIO PCA dim K_A — 64 is the sweet spot; both sides worse
MOTION PCA dim K_{MV} — no effect (flat)
GT
GT×other
AVTR-1 [unseeded]
AVTR-1 [seed0]
AVTR-1 [seed1]
DyStream [unseeded]
DyStream [seed0]
DyStream [seed1]
AvatarForcing [unseeded]
AvatarForcing [seed0]
AvatarForcing [seed1]
SoulX Lite
SoulX Pro
Ditto
FLOAT
AUDIO PCA dim K_A
32
default 64
96
MOTION PCA dim K_{MV}
16
default 32
64
−0.5
0.0
0.5
1.0
1.5
net ×1e4 (95% identity CI)
net ×1e4 (95% identity CI)
Common mask, n≈177, 13 identities, seed-averaged dyadic. Narrow K_A inflates the controls; wide K_A drops GT and turns a talking-head control significant. The motion dim changes nothing.